

Problem 1 — Elevator Direction

A building has floors numbered **1 to 30**. Read the **current floor** the elevator is on and the **target floor** a passenger wants. Decide what the elevator should do and how far it must travel.

Your program must

- read `current` and `target` as whole numbers
- if either floor is below 1 or above 30, print `Invalid floor` and stop deciding
- otherwise decide:
 - target higher → `Going up N floors`
 - target lower → `Going down N floors`
 - same floor → `You are already on this floor`
- `N` is how many floors it travels (the difference between the two floors)

Sample runs

Current floor: 3	Current floor: 20	Current floor: 7	Current floor: 35
Target floor: 12	Target floor: 5	Target floor: 7	Target floor: 4
Going up 9 floors	Going down 15 floors	You are already on this floor	Invalid floor

Hint: check the range first with `or`. For the distance, subtract the smaller floor from the larger so `N` is never negative — or compute it inside each branch.



Write your program here

```
current = int(input("Current floor:"))
target = int(input("Target floor:"))

if current < 1 or current > 30 or target < 1 or target > 30:
    print("Invalid floor")

elif current > target:
    print("Going down", current - target, "floor")

elif current < target:
    print("Going up", target - current, "floor")

else:
    print("You are already on this floor")
```

```
... Current floor:6
    Target floor:14
    You are already on this floor
```

int## Problem 2 — Cinema Ticket Price

A cinema sets ticket prices by age, then adds a weekend surcharge. The **base** price is **120 baht**. A **child** (age under 12) pays **half** the base. A **senior** (age 60 or over) pays **80 baht**. Everyone else pays the full 120. On a **weekend**, add **30 baht** to whatever the price is.

Your program must

- read the passenger's `age` (whole number)
- read whether it is a weekend as `y` or `n`
- decide the age-based price using `if` / `elif` / `else`
- if it is a weekend, add 30 to that price
- print `Ticket price: X baht`

Sample runs

Age: 10	Age: 70	Age: 30
Weekend (y/n): n	Weekend (y/n): y	Weekend (y/n): n
Ticket price: 60 baht	Ticket price: 110 baht	Ticket price: 120 baht

Hint: work out the age price first and store it in a variable. Then add the surcharge in a **separate** `if` that checks the weekend answer — don't mix the two decisions together.

```
 # Write your program here

age = int(input("Age:"))
weekend = input("Weekend (y/n)")

if age < 0 or age > 100:
    print("Invaild Error")

if age < 12:
    price = 60
elif age >= 60:
    price = 80
else:
    price = 120

if weekend == "y":
    price += 30

print ("Ticket price:", price, "baht")
```

Problem 3 — ATM Withdrawal

An ATM only dispenses cash in **multiples of 100 baht**. It refuses a withdrawal if the amount is **not** a multiple of 100, if it is **more than the balance**, or if it is **more than the 20000 daily limit**. Read the `balance` and the requested `amount`, then respond.

Your program must

- if the amount is not a multiple of 100 (`amount % 100 != 0`) → `Amount must be a multiple of 100`
- else if the amount is more than the balance → `Insufficient funds`
- else if the amount is more than 20000 → `Over daily limit`
- otherwise print `Dispensing X baht. New balance: Y`
- check the reasons **in this order** and print only one message (use `elif`)

Sample runs

Balance: 5000 Amount: 250 Amount must be a multiple of 100	Balance: 1000 Amount: 2000 Insufficient funds	Balance: 50000 Amount: 25000 Over daily limit	Balance: 5000 Amount: 3000 Dispensing 3000 baht. New balance: 2000
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Hint: the `%` operator gives the remainder; a multiple of 100 leaves remainder 0. Because you use `elif`, the first matching reason wins — so the order of your checks is part of the answer.

```
▶ balance = int(input())
  amount = int(input())

  if amount % 100 != 0:
      print("Amount must be a multiple of 100")
  elif amount > balance:
      print("Insufficient funds")
  elif amount > 20000:
      print("Over daily limit")
  else:
      new_balance = balance - amount
      print(f"Dispensing {amount} baht. New balance: {new_balance}")

... 1000
    2000
    Insufficient funds
```

Problem 4 — Electricity Bill

Monthly electricity is billed in **tiers** by the number of units used: the **first 150 units** cost **3 baht each**; the **next 250 units** (units 151–400) cost **4 baht each**; every unit **above 400** costs **5 baht each**. Read the number of `units` used and print the total bill.

Your program must

- read `units` (a whole number, 0 or more)
- use `if / elif / else` to find which tier the usage reaches
- charge each tier at its own rate — only the units that fall inside that tier
- print `Total bill: X baht`

Sample runs

Units: 100	Units: 300	Units: 500
Total bill: 300 baht	Total bill: 1050 baht	Total bill: 1950 baht

Hint: write the total for each band first — up to 150 → `units*3`; 151–400 → `150*3 + (units-150)*4`; above 400 → `150*3 + 250*4 + (units-400)*5`. Then use `if/elif/else` to pick the right one.



Write your program here

```
units = int(input())

if units <= 150:
    bill = units * 3
elif units <= 400:
    bill = (150 * 3) + ((units - 150) * 4)
else:
    bill = (150 * 3) + (250 * 4) + ((units - 400) * 5)

print(f"Total bill: {bill} baht")
```

```
... 300
    Total bill: 1050 baht
```